# Ruiyang Hao

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## Education

**Department of Automation, Tsinghua University, Beijing, China** | **2020.9-2023.7**
Electronic Engineering (MPhil) | GPA **4.00/4.00**
Supervisor: Prof. Biqing Huang | Ranking **1/159**

**Shenyuan Honors Collage, Beihang University, Beijing, China** | **2016.9-2020.6**
Electronic Engineering (BEng) | GPA **3.83/4.00**
Supervisor: Prof. Fei Tao | Ranking **1/51**

## Research Experience

### Research Assistant | AIR, Tsinghua University | Supervisor: Prof. Zaiqing Nie | 2023.7-now

- **Research Topic**: ***Vision Perception & Planning & Simulation for Autonomous Driving***
- **Specific Research Projects**:

✓ **Roadside Cooperative Perception Datasets and Benchmark (Lead contributor)**

***Objective***: Towards high-level autonomous driving, **R**oadside **Coo**perative **Per**ception (RCooper) task is proposed to realize more practical area-coverage roadside perception for restricted traffic areas, which can effectively extend sensing range and reduce blind spots.

***Contributions***:

1. Lead to construct the **first real-world, large-scale Roadside Cooperative Perception dataset** to bloom the research on practical roadside cooperative perception, including detection and tracking.
2. Construct the **roadside cooperative perception benchmark**, reimplementing SOTA cooperative methods,
3. Comprehensively analyze experimental results, demonstrating the effectiveness of roadside cooperation perception and the direction of further research.

✓ **New Open-Loop Evaluation Benchmarks for End-to-End Autonomous Driving (Primary contributor)**

***Objective***: To bridge the gap between open-loop vs closed-loop evaluation in End-to-End Autonomous Driving.

***Contributions***:

1. Comprehensively reveal the limitation of current L2 open-loop evaluation for end-to-end AD.
2. Propose **new open-loop evaluation benchmarks including four improvements** for the end-to-end autonomous driving planning task.
3. **Reevaluate existing SOTA methods using constructed benchmark**, demonstrating that proposed approach mitigates the problems associated with traditional open-loop evaluations.
4. Reimplement nuPlan devkit to **explore the relationship between open-loop and closed-loop benchmarks**, showing that a well-designed open-loop benchmark can serve as an effective rapid test for end-to-end autonomous driving.

✓ **More Realistic End-to-End Autonomous Driving Simulator (Primary contributor)**

***Overview***: Now I'm trying to improve the **diversity and complexity of driving agents** in an End-to-end Autonomous Driving Simulator, to enable the simulator to simulate more realistic driving scenarios.

✓ **VLM Decider for Autonomous Driving (Primary contributor)**

***Overview***: Now I'm also trying to construct a more **intelligent driving decider** leveraging the interpretability and interactivity of ***VLM***.

## Graduate RA | DA, Tsinghua University | Supervisor: Prof. Biqing Huang | 2020.2-2023.7

- **Research Topic:** ***Vision Perception for Industrial Defect Inspection***
- **Research Overview**: Visual detection and recognition for surface defects of industrial products plays a vital role in product quality control. Deep learning techniques have become powerful tools for this task, but there are still some challenges for actual industrial scenarios: 1) **the long-tail data distribution**; 2) the **lack of high-quality labeled data**; 3) **feature extraction for complex industrial defect**. Specific projects below show how I address mentioned issues.
- **Specific Research Projects**:

✓ **Glass defect Recognition based on Metric Learning and Long-Tail Learning (Lead contributor)**

***Contributions***:

1. Employ an adaptive threshold segmentation method to carry out high-recall and high-efficiency preliminary screening of glass defects.
2. Design a **metric learning task to extract robust embedding features** of glass defects under the condition of limited samples.
3. Construct an **ensemble learning framework with multi-expert classifiers** to recognize the similar defects class, where diversified loss functions are designed to address the long-tail data distribution.
4. Experimental results show that the defect recognition accuracy achieves 94% on the real-world dataset.

✓ **Unsupervised Segmentation of Industrial Surface Defects (Primary contributor)**

***Contributions***:

1. Design a **lightweight network structure with depth-wise convolution** for industrial deployment.
2. Employ **spital/channel-level attention** modules and **multi-task strategies** to improve the segmentation performance of defect edges.
3. Propose a **self-supervised learning strategy with homologous enhancement** to solve the defect segmentation problem without annotated information.
4. Experimental results show that proposed methods achieve 98.40% AUC in two typical industrial datasets.

✓ **Accurate Defect Inspection for Steel Surface Defects (Lead contributor)**

***Contributions***:

1. Employ a **deformable and atrous convolution structure** to improve feature extraction of multi-shape defects.
2. Leverage a **balanced feature pyramid with richer feature fusion** to improve feature extraction of multi-scale defects.
3. Experimental results show that proposed methods achieve 80.5% AP on steel surface defect datasets NEU-DET. And comprehensive ablation study is performed to show the model effects.

## Engineering Intern | ZCREATE Technology | Supervisor: Prof. Biqing Huang | 2022.3-2022.6

- **Research Topic**: ***Vision Perception for Driver Dangerous Driving Behavior Recognition***
- **Research Project**: **Dangerous Driving Behavior Recognition from the RSU view (Primary contributor)**
- **Contributions**:

1. Leverage the efficient YOLO algorithm to detect the local area around the driver.
2. Construct a **self-attention module for complex feature extraction** to extract features regarding the relationship between the drivers and their surroundings.
3. Implement a **class-rebalance strategy** to address the long-tail issue in dangerous driving behavior recognition task.
4. Experimental results show that proposed methods achieve 97% accuracy.

### Research Intern | EIG, SenseTime Research | Supervisor: Dr. Jiang Wu | 2022.6-2022.10

- **Research Topic**: ***Vision Perception for UAV Remote Sensing***
- **Research Project**: **Semantic segmentation for narrow objects from the UAV view (Lead contributor)**
- **Contributions**:

1. Leverage the **parallel multi-resolution branch and multi-level feature fusion** capabilities of HRNet to effectively improves the segmentation performance of small targets.
2. Construct a **multi-head multi-task decoder** to enhance the edge segmentation and direction perception.
3. Propose the **multi-modality integration** incorporating gradient information from the depth map to refine the perception of depth information.
4. Implement a **post-processing method considering continuity** to improve the performance in narrow objects, where the detected orientation of the principal normal vector is utilized to fix discontinuous targets.
5. Experimental results show a significant increase of over 30% in foreground pixel recall for UAV segmentation.

### Undergraduate RA | SIMRG, Beihang University | Supervisor: Prof. Fei Tao | 2019.2-2020.3

- **Research Topic**: ***Evolutionary Computation in Industry***
- **Research Project:** **Dual Diversities-aware industrial service supply-demand optimization (Lead contributor)**
- **Contributions**:

1. Highlight the type diversity and quantity variation diversity of manufacturing service supply and demand are essential for industrial service supply-demand matching.
2. Establish a **classification hypernetwork model** to describe the matching relationship between supply and demand of manufacturing services and their quantity relationship,
3. Propose a **utility optimization model** considering the type diversity and quantity diversity of supply and demand.
4. Design a **knowledge-guided NSGA-II multi-objective optimization algorithm** to solve the optimization model, and the effectiveness of the model and algorithm are demonstrated by comprehensive experiments.

## Publications

- **Ruiyang Hao**, B. Lu, Y. Cheng, X. Li, & Biqing Huang† (2021). A Steel Surface Defect Inspection Approach towards Smart Industrial Monitoring. ***Journal of Intelligent Manufacturing***, 32: 1833-1843. (*Link*, IF: 8.3, 108 citations)
- **Ruiyang Hao**, Ying Cheng†, Y. Zhang, & Fei Tao (2021). Manufacturing Service Supply-Demand Optimization with Dual Diversities for Industrial Internet Platforms. ***Computers & Industrial Engineering***, 156: 107237. (*Link*, IF: 7.9, 27 citations)
- R. Xu*, **Ruiyang Hao***, & Biqing Huang† (2022). Efficient surface defect detection using self-supervised learning strategy and segmentation. ***Advanced Engineering Informatics***, 52: 101566. (***co-first author**, *Link*, IF: 8.8, 49 citations)
- **Ruiyang Hao***, S. Fan*, Y. Dai, Z. Zhang, C. Li, Y. Wang, H. Yu, W. Yang, J. Yuan, & Zaiqing Nie† (2024). RCooper: A Real-world Large-scale Dataset for Roadside Cooperative Perception. ***Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2024)***, 22347-22357. (*Link*, 7 citations)
- K. Xu, **Ruiyang Hao*,** Hang Zhao, Xinlong Wang, Zaiqing Nie†, Ya-Qin Zhang (2024). Bridging the Open vs. Closed Loop Gap: New Open-Loop Evaluation Benchmarks for End-to-End Autonomous Driving Planning. *Under Review*. (***co-first author**)

- J. Wang, X. Wang, **Ruiyang Hao**, H. Yin, Biqing Huang[†], X. Xu, & J. Liu (2024). Incremental Template Neighborhood Matching for 3D Anomaly Detection. ***Neurocomputing***, 581: 127483. (*Link*, IF: 6.0)
- Y. Wang, X. Wang, **Ruiyang Hao**, B. Lu, & Biqing Huang[†] (2024). Metal surface defect detection method based on improved cascade R-CNN. ***Journal of Computing & Information Science in Engineering***, 24(4): 041002. (*Link*, IF: 3.1)

Notes: * denotes equal contribution and [†] denotes corresponding author. Reported citations are recorded in Jun. 2024.

## Professional Services

- Journal Reviewer of *Advanced Engineering Informatics*, *Journal of Industrial Information Integration*, *Journal of Real-Time Image Processing*, *The Visual Computer*, etc. Since 2023.
- Secretary, Graduate Thesis Review Committee, Tsinghua University, 2023.
- Member, Computer Vision Foundation (CVF), since 2024.
- Program Committee Member, ECCV 2024 Workshop on Coop-Intelligence, 2024.

## Selected Awards

- **National Scholarship**, the Top Scholarship in China, in 2018, **top 1%** of undergraduates
- First-class Excellence Scholarship, awarded by Tsinghua University, in 2022, **top 2%** of graduate students
- First-class Excellence Scholarship, awarded by Tsinghua University, in 2021, **top 2%** of graduate students
- Outstanding graduate and the honored degree, awarded by Beihang University, in 2020, **top 5%** of undergraduates
- Meritorious Prize of Interdisciplinary Contest in Modeling, awarded by the American Consortium for Mathematics and Its Applications, in 2019 and 2018, **top 8%** of contestants